Seat No.

College of Industrial Technology

King Mongkut's University of Technology North Bangkok

Midterm Examination of Semester 1 Year 2025

Subject 031001113 WAVE MOTION AND SOUND Section 15-17

Date Thursday, August 7, 2025 Time 10.00 - 12.00

Examination Setter Name: Asst.Prof. Dr.Apichat Siriwitpreecha and Asst.Prof. Dr.Jintawat Tanamatayarat

Student Name..ID...Class...............No.............

Directions;

1. This examination paper has not been edited. If any errors are identified, students are advised to exercise discretion in addressing them.
2. The examination consists of 26 questions on 7 pages (including the cover page). The total score is 35 points.
3. Complete all questions in the answer booklet.
4. Books, documents and lecture notes are not allowed.
5. Electronic calculators are not allowed
6. Students must remain in the examination room for no less than 1 hour from the start of the examination.
7. Do not open or start the examination before permission is granted. Students must strictly follow the instructions of the examination.
8. Access to restroom facilities during the examination is not permitted, except in cases of emergency
9. Students are not allowed to take the examination paper or copy the examination paper out of the examination room. Otherwise, it will be considered cheating.
10. Usage of all communication devices during the examination is strictly prohibited.
11. Smart watches (electronic watches) are not permitted during the exam.

Cheating in examinations is considered a serious offense and carries the penalty of expulsion from student status.

This examination is part of the assessment of students' abilities as follows:		Section 1	Section 2
CLO 1	Do experiment and explain the simple harmonic motion of objects attached to spring ends and simple pendulums. Including calculating various related quantities	1-6, 13-17	2.1
CLO 2	Explain wave phenomena, types of waves, wave parameters, observe and explain various properties of waves, as well as calculate speed, frequency, wavelength, sound propagation, relation of various quantities of waves, sound level, sound quality, sound resonance in pipes, including observation and explanation of beat formation, standing waves, doppler effect. sound shock wave Including calculating various quantities related to and applying the knowledge of sound to everyday life.	7-12, 18-23	2.2,2.3

Session 1: Multiple-choice Question (23 questions) (1 mark / question)

Instruction: mark an X for a correct choice of each question on the answer sheet provided.

Answer sheet												
	a	b	c	d	e			a	b	c	d	e
1							13					
2							14					
3							15					
4							16					
5							17					
6							18					
7							19					
8							20					
9							21					
10							22					
11							23					
12												

$$F_s = -kx \quad f = \frac{1}{T} \quad \omega = 2\pi f \quad EPE = \frac{1}{2}kx^2$$

$$x = A\cos\omega t \quad v_x = -\underbrace{A\omega}_{v_{\max}}\sin\omega t \quad a_x = -\underbrace{A\omega^2}_{a_{\max}}\cos\omega t$$

$$T = 2\pi\sqrt{\frac{L}{g}} \quad T = 2\pi\sqrt{\frac{m}{k}} \quad v = f\lambda \quad v = \sqrt{\frac{T}{m/L}} \quad f_n = n\left(\frac{v}{2L}\right)$$

$$f_{\text{beat}} = \left|f_1 - f_2\right| \quad f_n = n\left(\frac{v}{4L}\right) \quad v = 331 + 0.6T \quad I = \frac{P}{A}$$

$$\beta = 10\log\left(\frac{I}{I_o}\right) \quad f' = \left(\frac{v + v_O}{v - v_S}\right)f \quad \sin\theta = \frac{v}{v_S}$$

$$g = 10\ m/s^2$$

1. What is the direction of force that causing simple harmonic motion?
   a. Always directed toward the maximum distance
   b. Always directed toward the equilibrium position
   c. Always directed toward the right
   d. Always directed toward the left
   e. It depends on the position of the object

2. If the object is moved in simple harmonic motion, where is the position of maximum velocity?
   a. At the equilibrium position
   b. At the initial position
   c. At the maximum distance on the left
   d. At the maximum distance on the right
   e. Halfway between the maximum distance and the equilibrium

3. A mass of 500 g is attached to spring which has spring constant 2 N/m. What is period of vibration?
   a. $\frac{1}{\pi}$ s
   b. $\frac{1}{\pi^2}$ s
   c. $\frac{1}{2\pi}$ s
   d. π s
   e. 2π s

4. If the object is released at the distance of 5 cm from the equilibrium position of spring with constant of 50 N/m, what is the kinetic energy at the position of 3 cm?
   a. 0.01 J
   b. 0.02 J
   c. 0.04 J
   d. 0.0225 J
   e. 0.0625 J

5. From question 4, what is the angular frequency if it has maximum velocity of 0.1 m/s?
   a. 0.50 rad/s
   b. 0.10 rad/s
   c. 1.50 rad/s
   d. 2.00 rad/s
   e. 2.50 rad/s

6. Determine the length of a simple pendulum that will swing back and forth in simple harmonic motion with a period of 0.5 s.
   a. 1.6 cm
   b. 3.2 cm
   c. 4.9 cm
   d. 5.5 cm
   e. 6.3 cm

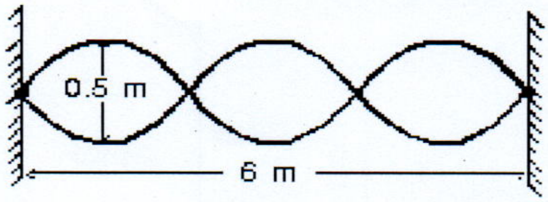
7. A string carries a sinusoidal wave with an amplitude of 2.0 cm and a frequency of 100 Hz. The maximum speed of any point on the string is:
   a. 2.0 m/s
   b. 4.0 m/s
   c. 6.3 m/s
   d. 13 m/s
   e. unknown (not enough information is given)

8. Two sound waves are traveling through the same medium. They have the same amplitude, wavelength, and direction of travel. If they are exactly out of phase, the type of interference they exhibit is:

a. fully constructive   b. fully destructive   c. indeterminate

d. partially constructive   e. partially destructive

9. A standing wave pattern is established in a string as shown. The wavelength of one of the traveling waves is:



a. 0.25 m   b. 0.5 m

c. 1 m   d. 2 m

e. 4 m

10. The smallest number of beats per second will be heard from which pair of tuning forks?

a. 20 and 23 Hz   b. 256 and 260 Hz   c. 534 and 540 Hz

d. 763 and 774 Hz   e. 10420 and 10422 Hz

11. What is the intensity level of sound wave which has intensity of 1×10^{-11} W/m^2?

a. 10 dB   b. 30 dB   c. 50 dB

d. 70 dB   e. 90 dB

12. How does the doppler effect occur?

a. Source is moving only
b. Observer is moving only
c. Source and observer move toward to each other
d. Source and observer move away to each other
e. All above are correct

Question 13-23

Read each statement below carefully.

Mark an X on the **option a** in the answer sheet if you think the statement is **TRUE.**

Mark an X on the **option e** in the answer sheet if you think the statement is **FALSE.**

13. In SHM motion, at the position of equilibrium position is zero acceleration.
14. The motion in SHM is straight line motion with constant acceleration.
15. The restoring force varies with the position of the object.
16. The total energy of SHM motion increases if the displacement of the object increases.
17. In simple pendulum, the frequency of the object depends on the length of string.
18. The speed of sound waves in the air depends on only its frequency.
19. If the tension of the string is increased, the fundamental frequency will increase.
20. Two sound waves are traveling through the same medium. They have the same amplitude, wavelength, and direction of travel. If the path difference is 5λ, the type of interference they exhibit is fully constructive.
21. If the sound level is increased by 10 db the intensity increases by a factor of 10,
22. Beats in sound occur when two waves of slightly different frequency interfere.
23. In Doppler effect, the source being in motion while the observer is at rest. As the source moves toward the observer, the wavelength appears shorter.

Session 2: Step-by-step explanation is required for this section.

(3 questions) (12 marks)

2.1 A spring can be stretched 0.15 m from equilibrium position by a force of 3 N. A 0.5 kg of mass attaches to this spring, it is then stretched 0.2 m from equilibrium position and released. Determine

a. Spring is constant (2 marks)

b. the amplitude of the motion (2 marks)

c. the maximum velocity and the maximum acceleration (2 marks)

2.2 A giant chorus of 1000 male vocalists is singing the same note, the measured sound intensity level is 120 dB Suddenly 999 vocalists stop, leaving one soloist. What is the sound intensity level? (3 marks)

2.3. A speaker emits sound waves with the frequency of 1000 Hz. An observer and the speaker move away to each other with speed of 2 m/s and 6 m/s, respectively. If the speed of sound in air is 344 m/s, what is the frequency at the observer? (3 marks)
